

Robert F. Archibald

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PhD-trained astrophysicist and Senior Data Scientist with 7+ years turning noisy, imperfect real-world data into validated, production-grade models. At Manulife, built and independently validated financial models against a third-party actuarial platform's reserve calculations to sub-1% error, repeatedly distinguishing genuine model improvements from metric noise under regulatory scrutiny. That same discipline, extracting reliable signal from messy, high-dimensional data and rigorously testing whether an apparent improvement is real, was the core of seven years of astrophysical research (25 peer-reviewed publications). Comfortable owning a problem end-to-end: exploratory analysis, feature engineering, model validation, and communicating findings clearly to technical and non-technical audiences.

Core Competencies

- **Model Validation, Uncertainty & Generalization:** Bias/variance assessment, real-improvement vs. metric-noise discrimination, production-scale scrutiny
- **Independent, Ambiguous Problem-Solving:** Self-directed research design, multi-telescope observing campaigns, model validation strategy
- **Experimental Design & Hypothesis Testing:** Cross-detector coincidence checks, false-alarm-probability thresholds, real-vs-artifact determination
- **Signal Extraction from Noisy, Real-World Data:** Confounding, data leakage, measurement variability, distribution shift; financial time series to astrophysical observations
- **Statistical & Quantitative Methods:** Python (scikit-learn, scipy, numpy, dask, pandas), machine learning, feature engineering, model selection, error analysis, time series analysis, stochastic simulation
- **Reproducible Production Python:** Version-controlled, unit-tested code; ETL and reconciliation across heterogeneous, imperfect data sources

Professional Experience

- **Manulife** **Montréal, Canada**
Senior Data Scientist 2019-2026
 - Built and independently validated ML models against a third-party actuarial platform's (AXIS) reserve calculations for conceptual soundness, bias, and generalization, using policyholder attributes as features; achieved sub-1% error against tens of billions of dollars of reserves while training on only 10% of the policy population, cutting compute cost by over 90%
 - Developed automated statistical comparison tooling to distinguish real model improvements from noise and isolate discrepancies attributable to model updates, assumption changes, or population shifts across 10s of GB of scenario results
 - Performed systematic error analysis and root-cause investigation on model disagreements, designing follow-up analyses to determine which of several competing explanations was correct rather than treating discrepancies as unexplained variance
 - Deployed actuarial tooling to an AKS cluster; produced clear technical documentation and presented validation findings to cross-functional teams and senior stakeholders
- **Manulife / ICON Consulting** **Toronto, Canada**
High Performance Computing Specialist 2018-2019
 - Designed and delivered a fully unit-tested data validation and ETL framework parsing and reconciling imperfect, heterogeneous data across COBOL files, SQL databases, CSVs, and Excel
 - Maintained Python Azure SDK tooling managing a ~2,000-VM scalable batch compute environment for large-scale quantitative workloads

- **University of Toronto** **Toronto, Canada**
Postdoctoral Fellow 2017-2018
 - Designed and implemented an automated transient-search pipeline over multi-year, 100s-of-GB NuSTAR X-ray datasets: flagged candidates via a false-alarm-probability threshold within a point-spread-function-sized region, then required coincident detections across NuSTAR's two co-aligned detectors to reject stray light and cosmic-ray artifacts
 - Applied time series analysis, signal processing, and statistical inference to validate the pipeline, recovering essentially all previously published magnetar bursts in the dataset and surfacing new transient candidates for follow-up

Education

- **McGill University** **Montréal, Canada**
PhD in Physics 2014-2017
Thesis: "X-ray Timing of Young Pulsars."
- **McGill University** **Montréal, Canada**
MSc in Physics 2011-2014
Thesis: "An Anti-glitch in a Magnetar."
- **Dalhousie University** **Halifax, Canada**
BSc in Physics 2007-2011
Thesis: "Factoring by Adiabatic Quantum Computation."

Publications

- 25 peer-reviewed articles in astrophysics, over 800 citations. Full list available on request.